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VEHICLE SEAT LONGITUDINAL ADJUSTMENT ASSEMBLY

Abstract

A method of assembling a release assembly for use in a vehicle seat. The method including attaching a spring to a coupling, the spring including a first end and a second end, wherein the coupling defines a first receptacle and a channel, and inserting the first end of the spring into the channel defined by the coupling. Inserting a portion of the coupling into a second receptacle defined by an actuation lever, the actuation lever fixed to a transverse member, and configured to pivot about an axis defined transverse member to engage a locking lever of the locking device to unlock the translatable rail from the fixed rail. The method further including rotating the coupling in a first rotational direction, relative to the second receptacle, to a first rotational position so that the second end of the spring engages the actuation lever.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to vehicle seats, including vehicle seats provided with a manual longitudinal adjustment assembly and a release assembly for the same.

BACKGROUND

[0002] A vehicle seat with a release assembly is described in German Utility Patent DE 203 13 952 U1.

SUMMARY

[0003] According to one embodiment, a vehicle seat is provided. The vehicle seat may include a first rail assembly, a first locking device, and a release assembly. The release assembly may include a handle, a transverse member, a first release lever, a first coupling, and a first spring. The first rail assembly may include a first fixed rail and a first translatable rail, the first fixed rail may be configured to be fixed to the vehicle floor and the first translatable rail may be configured to translate along the first fixed rail. The first locking device may be coupled to the first rail assembly and configured to selectively change between a locked state, in which the first locking device locks the first translatable rail to the first fixed rail, and an unlocked state in which the first locking device unlocks the first translatable rail from the first fixed rail. The transverse member of the release assembly may be fixed to the handle and the first actuation lever. The first coupling may include a flange and a barrel extending from the flange, the barrel may form a first receptacle and a channel. The first actuation lever may include a distal end portion, a proximal end portion, and a medial portion extending there between. The first actuation lever may further include a second receptacle extending from the proximal end portion. The distal end portion may be configured to engage a locking lever of the locking device to change the state of the locking device from the locked state to the unlocked state. The proximal portion may form a second receptacle that may be configured to receive at least a portion of the barrel. The first spring may include a fixed leg and a free leg. The fixed leg may be disposed in the channel defined by the barrel and the free leg of the first spring may be configured to bias the distal end portion of the release lever against the locking lever of the locking device.

[0004] According to another embodiment, a release assembly for use in a vehicle seat is provided. The vehicle seat may include a pair of rail assemblies and each rail assembly of the pair of rail assemblies may include a fixed rail, a translatable rail, and a locking device. The locking device may be configured to selectively lock the translatable rail to the fixed rail and unlock the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail. The release assembly may include a coupling, a spring, and an actuation assembly. The actuation assembly may include an actuation lever that may define a second receptacle, a transverse member, and a handle. The actuation lever and the handle may each be coupled to the transverse member. The coupling may include a barrel and a flange that may extend from the barrel. A portion of the coupling may define a channel and the barrel may define the first receptacle. The spring may include a first end and a second end. The channel defined by the coupling may be configured to receive the first end of the spring. At least a portion of the barrel may be configured to nest within the second receptacle defined by the actuation lever so that the second end of the spring lies against a portion of the actuation lever. In a pre-installed state, in which the coupling is not fixed relative to the translatable rail, the coupling may be in a first position with respect to the second receptacle. In an installed state, in which the coupling is fixed relative to the translatable rail, the coupling is in a second position with respect to the second receptacle. As an example, the second position may be different than the first position.

[0005] According to yet another embodiment, a method of assembling a release assembly for use in

a vehicle seat is provided. The vehicle seat may include a pair of rail assemblies and each rail assembly of the pair of rail assemblies may include a fixed rail, a translatable rail, and a locking device. The locking device may be configured to selectively lock the translatable rail to the fixed rail and unlock the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail. The method may include: attaching a spring to a coupling, the spring may include a first end and a second end, and the coupling may define a first receptacle and a channel. The attaching step may include inserting the first end of the spring into the channel defined by the coupling. The method may also include inserting a portion of the coupling into a second receptacle defined by an actuation lever. The actuation lever may be fixed to a transverse member and may be configured to pivot about an axis defined by the transverse member to engage a locking lever of the locking device to unlock the translatable rail from the fixed rail. The method may further include rotating the coupling in a first rotational direction relative to the second receptacle to a first rotational position so that the second end of the spring engages the actuation lever.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates a schematic diagram of an exemplary vehicle seat.

[0007] FIG. 2 illustrates an exploded view of a release assembly and a pair of rail assemblies configured for use in a vehicle seat.

[0008] FIG. 3 illustrates an exploded view of a portion of the release assembly.

[0009] FIG. 4 illustrates an exploded view of the portion of the release assembly illustrated in FIG. 3 and a portion of an actuation assembly.

[0010] FIG. 5A illustrates a perspective view of the portion of the release assembly illustrated in FIG. 4 in a delivery position.

[0011] FIG. 5B illustrates a perspective view of the portion of the release assembly illustrated in FIG. 4 in an installed state.

[0012] FIG. 6A illustrates a side plan view of the portion of the release assembly illustrated in FIG. 5A in a pre-installed state.

[0013] FIG. 6B illustrates a side plan view of the portion of the release assembly shown in FIG. 6A in the installed state.

[0014] FIG. 6C illustrates a perspective view of the release assembly shown in FIG. 6A in the installed state

[0015] FIG. 7 illustrates an exploded perspective view of another release assembly and a pair of rail assemblies configured for use in a vehicle seat.

[0016] FIG. 8 illustrates a perspective view of the release assembly illustrated in FIG. 7 as the release assembly is installed to a vehicle rail.

[0017] FIG. 9 illustrates an exploded perspective view of another release assembly and a pair of rail assemblies configured for use in a vehicle seat.

[0018] FIG. 10 illustrates a perspective view of one rail assembly of the pair of rail assemblies illustrated in FIG. 9.

[0019] FIG. 11 illustrates a perspective view of the release assembly illustrated in FIG. 9 installed to a rail assembly illustrated in FIG. 9.

[0020] FIG. 12 illustrates a perspective view of an exemplary rail assembly in an unlocked state.

DETAILED DESCRIPTION

[0021] Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional

details disclosed herein are not to be interpreted as limiting, but merely as a representative bases for teaching one skilled in the art to variously employ the embodiments. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures can be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical application. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

[0022] “A”, “an”, and “the” as used herein refers to both singular and plural referents unless the context clearly dictates otherwise. By way of example, “a processor” programmed to perform various functions refers to one processor programmed to perform each and every function, or more than one processor collectively programmed to perform each of the various functions.

[0023] The term “substantially” or “about” may be used herein to describe disclosed or claimed embodiments. The term “substantially” or “about” may modify a value or relative characteristic disclosed or claimed in the present disclosure. In such instances, “substantially” or “about” may signify that the value or relative characteristic it modifies is within $\pm 0\%$, 0.1% , 0.5% , 1% , 2% , 3% , 4% , 5% or 10% of the value or relative characteristic.

[0024] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). The term “and/or” includes any and all combinations of one or more of the associated listed items.

[0025] Although the terms first, second, third, etc. may be used to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0026] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0027] Vehicle seats, in particular those vehicle seats that are manually adjustable to change the longitudinal position (e.g., in the fore-aft direction) of the seat include a release assembly that may be actuated to unlock the translatable rails from the fixed rails so that the user may slide the vehicle seat in the forward and rearward directions. Generally, the manufacturer of the vehicle seat may assemble the rail assemblies as separate components and marry the rail assemblies to one another by attaching the release assembly and one or more other assemblies, such as an upper frame assembly or cushion portion to form a seat sub-assembly or the complete seat. The manufacturer will then supply the seat sub-assembly or complete seat to an original equipment manufacturer

(OEM) or a higher tier manufacturer. The OEM will then incorporate the complete seat into the desired motor vehicle.

[0028] Under certain circumstances, the OEM or higher tier manufacturer may desire to assemble portions of the seat in their own facility prior to incorporating the seat into vehicles. As an example, the OEM may require the seat manufacturer to ship the rail assemblies and the release assembly as separate components and the OEM may then assemble the release assembly to the rail assemblies prior to incorporating the vehicle seat into the vehicle. Typically, assembling the release assembly to the rail assemblies requires placing the rail assemblies on floating fixtures and positioning the release assembly between each rail assembly and sliding the rail assemblies towards the release assembly to fix the release assembly to the each of the rail assemblies. In short, this requires placing each rail assembly, placing the release assembly, positioning each of the rail assemblies onto the release assembly, and fixing the release assembly to each of the rail assemblies. This process has been found to be relatively complicated and time consuming.

[0029] The disclosure provided herein attempts to address the above-mentioned shortcomings.

[0030] FIG. 1 illustrates a schematic illustration of an exemplary vehicle seat **100** provided with a cushion portion **102** and a backrest **104**. A fitting or recliner **108** may connect the backrest **104** to one or more portions of the cushion portion **102** such as a seat cushion **106**. The recliner **108** may be configured to rotate about an axis A to adjust the inclination of the backrest **104** as represented by the bi-directional arrow disposed above the backrest **104**. The cushion portion **102** may include one or more (e.g., two) rail assemblies **110** that may be fixed to a vehicle floor **116**.

[0031] The rail assemblies **110** may each include a fixed rail **112** and a translatable rail **114** that may be configured to translate along the fixed rail **112** to adjust the cushion portion **102** and backrest **104** in a longitudinal direction as represented by the x-axis. The cushion portion **102** may include an actuation assembly **118** and release assembly **120** and at least one of the rail assemblies **110** may include a locking device **132** (FIG. 3) that may be actuated in response to a force applied to the release assembly **120** in one or more directions (e.g., vertical direction, as represented by the z-axis).

[0032] FIG. 2 illustrates a perspective view of the release assembly **120**, including the actuation assembly **118**, that is disposed above a pair of rail assemblies **110**, **110'**. FIG. 3 illustrates a perspective-exploded view of a portion of the release assembly **120**. For purposes of clarity, the rail assemblies **110** are depicted without brackets (except for mounting bracket **128**) generally fixed to the fixed rails **112** and translatable rails **114**. And reference numeral **110** generally refers to both rail assemblies as well as the left-hand rail assembly **110**. Reference numeral **110'** refers to the right-hand rail assembly **110'**. As will be described in greater detail below, the actuation assembly **118** may be configured as a modular unit or sub-assembly that may be shipped to a higher tier manufacturer or OEM so that the higher tier manufacturer or OEM may assemble the actuation assembly **118** to the rail assemblies **110** in a relatively easy and cost-efficient manner.

[0033] The actuation assembly **118** may include a handle **122**, a transverse member **124**, and one or more release levers or actuation levers **126**. The transverse member **124** may be formed by a metal tube or other cylindrical shaped member that may be fixed (e.g., welded) to the handle **122** and each of the actuation levers **126**. Note, one or more of the handle **122**, release levers **126**, and transverse member **124** may be integrally formed or connected to one another as opposed to being separate components as depicted in FIG. 2. As a force is applied to the handle **122**, the transverse member **124** may rotate about a rotational axis RA to pivot the actuation lever **126** to unlock the translatable rails **114**, **114'** from the fixed rails **112**, **112'**.

[0034] One or more of the rail assemblies **110** may include a mounting bracket **128** that may be configured to receive and fix portions of the release assembly **120** to the rail assemblies **110**. In one or more embodiments, the mounting bracket **128** may be fixed to a portion of each of the translatable rails **114**, **114'** such as to a top wall or surface of each translatable rail **114**, **114'**. In one or more embodiments, the mounting bracket **128** may include a main body **129** and an edge portion

131 that may extend from the main body **129** so that the edge portion is disposed between the left-hand rail assembly **110** and the right-hand rail assembly **110'**.

[0035] The mounting bracket **128** may include one or more features that are configured to retain or fix the housing or coupling **130** of the release assembly **120** to each of the rail assemblies **110**, **110'**. As an example, the edge portion **131** may define or form a recess **133** that may have a curved shape so that when the release assembly **120** is fixed to each of the rail assemblies **110**, **110'**, a bottom portion of the barrel **152** (FIG. 3) of the housing or coupling **130** nests within the recess **133**. Because of the recess **130**, the release assembly **120** is positioned so that distal end portions **170** of the actuation levers **126** engage portions of the locking device **132**.

[0036] Each of the rail assemblies **110**, **110'** may include a locking device **132** that may be disposed rearward of the mounting bracket **128**. An exemplary locking device is disclosed in German Utility Patent DE 203 13 952 U1 which is hereby incorporated by reference.

[0037] Referring now to FIG. 12, the locking device **132** may include one or more (e.g., three) pins **138** that may extend through the top wall of the translatable rail **114** and engage holes or slots formed in the fixed rail when the locking device **132** is in the locked state. The locking device **132** may include one or more levers such as a locking lever **136** and a pin lever **136** that may cooperate with one another to move (e.g., raise) the pins **138** so that the pins **138** disengage from the fixed rail **112**, **112'** and change the state of the locking device **132** from the locked state to the unlocked state to permit the translatable rails **114**, **114'** to translate along the fixed rails **112**, **112'**. When the release assembly **120** is in the installed position, the actuation lever **126** may engage or lie along a portion of the locking lever **136**.

[0038] Referring back to FIG. 2 and FIG. 3, the release assembly **120** may include the actuation assembly **118**, one or more spring housings or couplings **130**, and one or more springs **142**. The coupling **130** may include a barrel **152**, one or more flanges **148**, **150**, an annular ring **156**, and a protrusion **158**. Note in one or more embodiments, the coupling is **130** may be an injection molded component formed of one or more polymeric materials including thermoplastics such as Polypropylene, such as a glass-filled Polypropylene (Celstran PP-GF30-0453 P10/0), or Polyamide 66 (PA66) or another suitable material, as required and one or more of the components that form the coupling may be integrally formed with one another. As an example, the barrel **152** may include the annular ring **156** and the annular ring **156** may include the protrusion **158**. As will be described in greater detail below, the coupling **130** may be configured to form a pre-assembly or sub-assembly that may be supplied to another manufacturer or OEM and assembled to the rail assemblies **110** in a relatively efficient manner.

[0039] In one or more embodiments, the barrel **152** may have an elongated body, that has a length that extends in the longitudinal direction (e.g., the y-axis) that is greater than the width (e.g., extending in the x-direction) of the barrel **152**. The one or more flanges may include the first flange **148** and the second flange **150** that may each extend in the longitudinal direction (e.g., as represented by the x-axis). The first flange **148**, the second flange **150**, or both may each define one or more attachment apertures **154** that may be configured to receive one or more fasteners **134** that may thread into or engage one or more apertures **202** defined by the mounting bracket **128**.

[0040] The annular ring **156** may extend from the **152** in the axial direction (e.g., y-direction) and the outer diameter, inner diameter, or both of the annular ring **156** may be greater than the inner diameter, outer diameter, or both the inner and outer of the barrel **152**. As an example, the inner periphery of the annular ring **156** may form or define a spring receptacle **160**. An inner periphery of the barrel **152** may form transverse member receptacle **162** that may be configured to receive an end portion **166** (FIG. 4) of the transverse member **124**.

[0041] The release assembly **120** may include one or more springs **142** that may be configured to set the rotational position of the coupling **130** relative to the actuation lever (e.g., in a pre-installed state or prior to assembling the release assembly **120** to the pair of rails **110**) and apply a biasing force to the actuation lever **126** (e.g., in an installed state or when the release assembly **120** is

assembled to the pair of rails **110**). The spring **142** may be a torsion spring provided with a first end or fixed end **144**, a second end or free end **146**, and one or more coils that may extend between the fixed end **144** and the free end **146**. The free end **146** of the spring **142** is shown in a free state, in which the free end **146** does not engage the actuation lever **126**, and an engaged state or tensioned state (as represented by hidden lines in FIG. 3), in which the free end **146** engages the actuation lever **126**. As an example, the barrel **152** may define an aperture or channel **164** that may receive the fixed end **144** of the spring and the spring receptacle **160** may be configured or sized to receive the one or more coils extending between the free end **146** and the fixed end **144**.

[0042] FIG. 4 illustrates a perspective view of a portion of the actuation assembly **118**, including the actuation lever **126** and the transverse member **124**, being assembled to the coupling or spring housing **130** provided with the spring **142**. As described above, the spring receptacle **160** may receive the one or more coils **168** of the spring **142** after the fixed end **144** of the spring **142** is inserted into the channel **164** of the coupling **130**. The actuation lever **126** may include a distal end **170**, a proximal end **174**, and a medial portion **172** extending there between. A semi-annular ring **184** may extend from the proximal end **174** or the semi-annular ring **184** may form a portion of the proximal end **174**. The semi-annular ring **184** may form a housing or coupling receptacle **182** that may be configured to receive at least a portion of the coupling **130** (e.g., a portion of the annular ring **156**).

[0043] In one or more embodiments, the semi-annular ring **184** may include a first portion **194** and a second portion **196**. The first portion **194** may extend from the proximal end **174** and terminate at an edge **188** and the second portion may extend from the proximal end **174** and terminate at a flange **186** that may extend beyond the edge **188**. As will be described in greater detail below, the flange **186** may be configured to lock or fix the housing or coupling **130** with respect to the axial direction (e.g., as represented by the y-axis).

[0044] As an example, one or more portions of the actuation lever **126** may be formed by a metal stamping provided with a first leg **176** and a second leg **178** that may be arranged substantially orthogonally to one another. It should be understood that the actuation lever may be formed by another material (e.g., plastic) or another process (e.g., injection molding, additive manufacturing), as required. The first leg **176** may form a bottom portion or base **180** of the coupling receptacle **182** and the semi-annular ring **184** may extend from a portion of the base **180** that defines a bottom **190** of the coupling housing **182**. The base **180** may define an aperture **192** and an end portion **166** of the transverse member **124** may extend through the aperture **192**. The coupling or housing **130** may be slid (e.g., along the directional arrow D1) and the transverse member receptacle **162** (FIG. 3) may receive the end portion **166** of the transverse member **124**.

[0045] FIG. 5A illustrates the housing or coupling **130** being assembled to a portion of the actuation assembly **118** and FIG. 5B illustrates the housing or coupling **130** installed to the transverse member **124** and the actuation lever **126**. As the housing or coupling **130** (provided with the spring **142**) is assembled to the transverse member **124** and the coupling receptacle **182**, an operator (e.g., human or machine) may manipulate the coupling **130** to rotate the coupling **130** in a first rotational direction, as represented by directional arrow R1, and move or slide the coupling **130** in the axial direction so that the protrusion **158** moves past the flange **186** in the axial direction (e.g., the represented by y-axis). The position of the housing or coupling **130** relative to the actuation lever **126** may be referred to as the first rotational position P1. As an example, as the coupling **130** is rotated and slid into the first rotational position P1, the free end **146** of the spring **142** may lie against a portion (e.g., the second leg **178**) of the actuation lever **126** (FIG. 6A) so that each end of the spring **142** is substantially fixed; the free end **146** lies against the actuation lever **126** and the fixed end **144** is disposed in the channel **164**.

[0046] Once in position, the operator may release the coupling **130** and torque applied by the spring **142** may rotate the coupling in the second rotational direction R2, as represented by directional arrow R2. As an example, the spring **142** may rotate the coupling **130** so that the

coupling is in a second rotational position P2 with respect to the actuation lever so that a surface **198** of the protrusion **158** that faces the edge **188** is within a predetermined distance (e.g. radial angle) of the edge **188**. As another example, when the coupling is in the second rotational position P2, the surface **198** may contact or lie along the edge **188**.

[0047] FIG. 6A illustrates a side-plan view of the housing or coupling **130**, the spring **142**, and portions of the actuation assembly **118** in the installed state. For purposes of clarity, the fixed rail **112**, translatable rail **114**, and the mounting bracket **128** are not illustrated. FIG. 6C illustrates a perspective view of the housing or coupling **130** and portions of the actuation assembly **118** in the installed state. As mentioned above, when the housing or coupling **130** is seated within the receptacle **182**, the free end **146** of the spring **142** lies along one or more portions of the actuation lever **126**, and the coupling **130** is in the second rotational position P2. And as the housing or coupling **130** is fixed to the rail assembly **110**, **110'** (e.g., the mounting bracket **128**) the housing or coupling **130** may rotate in the first rotational direction R1 (FIG. 6B) to a third rotational position P3, in which the surface **198** is spaced apart from the edge **188** by a predetermined distance (e.g., angular distance θ).

[0048] As an example, the predetermined distance may be an angular distance θ measured between two lines each extending from the rotational axis RA of the transverse member **124** to a point on the surface **198** and a point on the edge **188**. Alternatively or additionally, the predetermined distance may be a straight line distance between a point on the surface **198** and a point on the edge **188**. The straight line distance may range between 2.5 mm and 15 mm.

[0049] FIG. 6C illustrates a partial perspective of the right-hand rail assembly **110'** and the release assembly **120** assembled (e.g., in the installed state) to the right-hand rail assembly **110'**. As mentioned above, the locking device **132** includes the locking lever **136** and a pin lever **140** coupled to the locking lever **136**. The locking device **132** may further include a torsion spring **204** that biases the pin lever **140** towards the top surface of the translatable rail **114'** so that one or more (e.g., two) pins **138** are in a deployed position and engage the fixed rail **112'**. And, in response to a force applied to the handle **122** (FIG. 2), the transverse member **124** rotates relative to the housing or coupling **130** and the actuation lever **126** pivots and moves (e.g., pivots) the locking lever **136** towards the fixed rail **112'**.

[0050] In response to actuation of the locking lever **136**, the pin lever **140** may be actuated to lift the one or more pins **138** to unlock the translatable rail **114'** from the fixed rail **112'**. As the coupling **130** is fixed to the mounting bracket **128**, the coupling **130** may rotate in the first rotational direction (FIG. 6B) to the third rotational position P3 so that the free end **146** of the spring **142** applies a force to the actuation lever **126** so that the actuation lever **126** is biased against the locking lever **136**. The force of the spring **142** is insufficient to overcome the force of the pin lever spring **204**, so that without an external force being applied to the handle **122** (e.g., by a user), the locking device remains in a locked state, in which the pins **138** lock the translatable rail **114'** to the fixed rail **112'**.

[0051] FIG. 7 illustrates a perspective view of a release assembly **206** as the release assembly **206** is assembled the pair of rail assemblies **110**, **110'**. The release assembly **206** may include a another housing or coupling **200** provided with a hook **208** that may extend from either the first flange **148** or the second flange **150**. The hook **208** may form a gap that is configured to receive a portion of the mounting bracket **128** as the release assembly is fixed or assembled to the rail assemblies **110**, **110'**. In addition to the hook **208**, one or more fasteners **134** may be used to extend through the mounting aperture **154** (FIG. 3) and thread into the one or more apertures **202** defined by the mounting bracket **128**.

[0052] FIG. 8 illustrates another perspective view of the release assembly **206** as the release assembly is assembled to the right-hand rail assembly **110'**. Here, the hook **208** of the coupling **200** extends from the second flange **150** and not the first flange **148** as illustrated in FIG. 7. In one or more embodiments, the hook **208** may include a first arm **210** and a second arm **212** that may be

arranged orthogonal to the first arm **210**. As shown, the second arm **212**, represented by hidden lines, is disposed below or on the bottom side of the mounting bracket **128**. The second arm **212** of the hook **208** may be arranged substantially parallel to the second flange **150** and the second arm **212** and the second flange **150** may form a gap **G** that is configured to receive a portion (e.g., part of the edge portion **131**) of the mounting bracket **128**. The hook **208** may also act as a locating feature to assist an operator in assembling the release assembly **206** to the rails **110**, **110'**. After the hook **208** engages the mounting bracket **128**, the first flange **148** may be pressed down to lie along the mounting bracket **128** and one or more fasteners **134** (FIG. 7) may be driven into the mounting aperture **154** and into the aperture defined by the mounting bracket **128**.

[0053] FIG. 9 illustrates a perspective view of a release assembly **214** as the release assembly **214** is assembled the pair of rail assemblies **110**, **110'**. Each of the rail assemblies **110**, **110'** may include a mounting bracket **218** according to another embodiment and the release assembly **214** includes a housing or coupling **216** according to another embodiment. FIG. 10 illustrates a perspective view of the mounting bracket **218** fixed to the translatable rail and FIG. 11 illustrates a perspective view of the coupling **216** fixed to the mounting bracket **218**. As will be described in greater detail below, the configuration illustrated in FIGS. 9-11 may facilitate fixing the release assembly **214** without the use of fasteners.

[0054] In one or more embodiments, the housing or coupling **216** may include a main body **228** and a semi-round portion that may be formed by a barrel **224**. The main body **228** may define one or more attachment slots **222** that may be disposed on either side (e.g., with respect to the longitudinal direction) of the transverse member **124** of the actuation assembly **118**. The mounting bracket **218** may be similar to the mounting bracket **128** described above, in that the mounting bracket **218** includes the main body **129** and the edge portion **131** that defines or forms the recess **133**. The recess **133** may be configured to receive a portion of the barrel **224** disposed below the main body **228** of the coupling **216**.

[0055] The mounting bracket **218** may further include one or more locking protrusions **220** that may extend in a vertical direction from the edge portion **131**. The one or more locking protrusions **220** may include one or more retention members such as one or more serrations **226** that may be configured to engage an inner periphery of each of the slots **222** to fix the coupling **216** to the mounting bracket **218**. The locking protrusions **220** and the slots **222** may be collectively configured to fix the coupling **216** in the longitudinal direction (e.g., represented by the x-axis) and the axial or cross-car direction (e.g., represented by the y-axis).

[0056] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, to the extent any embodiments are described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics, these embodiments are not outside the scope of the disclosure and can be desirable for particular applications.

[0057] The following is a list of reference numbers shown in the Figures. However, it should be understood that the use of these terms is for illustrative purposes only with respect to one

embodiment. And, use of reference numbers correlating a certain term that is both illustrated in the Figures and present in the claims is not intended to limit the claims to only cover the illustrated embodiment.

PARTS LIST

[0058] **100** vehicle seat [0059] **102** cushion portion [0060] **104** back rest [0061] **106** seat cushion [0062] **108** recliner [0063] **110** rail assemblies [0064] **112** fixed rail [0065] **114** translatable rail [0066] **116** vehicle floor [0067] **118** actuation assembly [0068] **120** release assembly [0069] **122** handle [0070] **124** transverse member [0071] **126** actuation lever [0072] **128** mounting bracket [0073] **129** main body mounting bracket [0074] **130** spring housing, coupling [0075] **131** edge portion mounting bracket [0076] **132** locking device [0077] **133** recess of mounting bracket [0078] **134** fasteners [0079] **136** locking lever [0080] **138** pins [0081] **140** pin lever [0082] **142** spring [0083] **144** first end, fixed end of spring [0084] **146** second end, free end of spring [0085] **148** first flange [0086] **150** second flange [0087] **152** barrel [0088] **154** mounting aperture [0089] **156** annular ring [0090] **158** protrusion [0091] **160** spring receptacle, first receptacle [0092] **162** transverse member receptacle, third receptacle [0093] **164** channel [0094] **166** end portion of transverse member [0095] **168** coils [0096] **170** distal end actuation lever [0097] **172** medial portion actuation lever [0098] **174** proximal end actuation lever [0099] **176** first leg [0100] **178** second leg [0101] **180** base [0102] **182** coupling receptacle [0103] **184** semi-annular ring [0104] **186** flange [0105] **188** edge [0106] **190** bottom of receptacle [0107] **192** aperture of base [0108] **194** first section [0109] **196** second section [0110] **198** surface of protrusion [0111] **200** coupling [0112] **202** aperture mounting bracket [0113] **204** pin lever spring [0114] **206** release assembly [0115] **208** hook [0116] **210** first leg [0117] **212** second leg [0118] **214** release assembly [0119] **216** coupling [0120] **218** mounting bracket [0121] **220** locking protrusion [0122] **222** slot [0123] **224** barrel [0124] **226** serrations [0125] **228** main body

Claims

1. A vehicle seat comprising: a first rail assembly including a first fixed rail and a first translatable rail, the first fixed rail configured to be fixed to a vehicle floor and the translatable rail configured to translate along the first fixed rail; a first locking device coupled to the first rail assembly and configured to selectively change between a locked state, in which the first locking device locks the first translatable rail to the first fixed rail, and an unlocked state in which the first locking device unlocks the first translatable rail from the first fixed rail; and a release assembly provided with, a handle including a transverse member and a handle portion extending from the transverse member, a first release lever including a distal end portion, a proximal end portion, and a medial portion extending there between, wherein the distal end portion is configured to engage a locking lever of the locking device, wherein the proximal portion includes a first receptacle and defines an aperture, wherein the aperture is configured to receive a portion of the transverse member, a first coupling including a flange and a barrel extending from the flange, the barrel defining a second receptacle and a channel, and a first spring including a fixed leg and a free leg, wherein the fixed leg is disposed in the channel and the free leg of the spring configured to bias the proximal end portion of the release lever against the locking lever of the locking device.
2. The vehicle seat of claim 1, wherein the first spring includes one or more coils, the one or more coils extending between the fixed leg and the free leg of the first spring and disposed in the second receptacle.
3. The vehicle seat of claim 1, wherein the flange of the first coupling is configured to be coupled to the first translatable rail so that the first coupling moves with the first translatable rail as the first translatable rail translates along the first fixed rail.
4. The vehicle seat of claim 1, further comprising: a mounting bracket including a main body and an edge portion, the main body fixed to the first translatable rail and the edge portion defining a

recess configured to receive a portion of the barrel of the first coupling.

5. The vehicle seat of claim 4, wherein the mounting bracket includes at least one protrusion provided with at least one retention member, wherein the flange of the first coupling defines at least one attachment aperture, wherein the at least one attachment aperture receives the at least one protrusion and the at least one retention member retains the first coupling to the mounting bracket.

6. The vehicle seat of claim 4, wherein the first coupling includes an arm extending from the flange, wherein the arm includes a first portion and a second portion, the first portion extending from the flange and the second portion extending from the first portion and parallel to the flange to form a gap, wherein the gap receives an edge portion of the mounting bracket.

7. The vehicle seat of claim 1, further comprising: a second rail assembly including a second fixed rail and a second translatable rail, the second fixed rail configured to be fixed to a vehicle floor and the second translatable rail configured to translate along the second fixed rail; and a second locking device coupled to the second rail assembly and configured to selectively change between a locked state, in which the second locking device locks the second translatable rail to the second fixed rail, and an unlocked state in which the second locking device unlocks the second translatable rail from the second fixed rail; a first mounting bracket including a main body and an edge region, wherein the main body of the first mounting bracket is fixed to the first translatable rail and the edge region of the first mounting bracket extends from the main body of the first mounting bracket and is disposed between the first rail assembly and the second rail assembly; and a second mounting bracket including another main body and another edge region, wherein the main body of the second mounting bracket is fixed to the second translatable rail and the edge region of the second mounting bracket extends from the main body of the second mounting bracket and is disposed between the first rail assembly and the second rail assembly, wherein the flange of the first coupling includes a first hook, wherein the release assembly is further provided a second coupling provided with a second hook, wherein as the release assembly is assembled to the first rail assembly and the second rail assembly, the first hook and the second hook are each configured to engage and pivot about the edge region of the first mounting bracket and the second mounting bracket, respectively.

8. A release assembly for use in a vehicle seat, the vehicle seat including a pair of rail assemblies, wherein each rail assembly of the pair of rail assemblies includes a fixed rail, a translatable rail, and a locking device, the locking device configured to selectively lock the translatable rail to the fixed rail and unlock the translatable rail from the fixed rail to permit translation of the translatable rail along the fixed rail, the release assembly comprising: a coupling including a barrel and a flange extending from the barrel, wherein a portion of the coupling defines a channel and the barrel defines a first receptacle; a spring including a first end and a second end, wherein the channel is configured to receive the first end of the spring; and an actuation assembly, the actuation assembly including, an actuation lever defining a second receptacle, a transverse member, and a handle, wherein the actuation lever and the handle are each coupled to the transverse member, wherein at least a portion of the barrel is configured to nest within the second receptacle defined by the actuation lever so that the second end of the spring lies against a portion of the actuation lever, and wherein in a pre-installed state, in which the coupling is not fixed relative to the translatable rail, the coupling is in a first position with respect to the second receptacle, and wherein in an installed state, in which the flange of the coupling is fixed relative to the translatable rail, the coupling is in a second position with respect to the second receptacle, the second position different than the first.

9. The release assembly of claim 8, wherein the actuation lever includes a proximal end, a distal end, and a semi-annular ring, wherein the distal end is configured to engage and displace a locking lever of the locking device to unlock the translatable rail from the fixed rail, wherein the semi-annular ring forms at least a portion of the second receptacle and extends from the proximal end of the actuation lever and terminates at an edge, wherein the barrel includes an annular ring and a protrusion, wherein the annular ring defines the first receptacle and the protrusion extends from an

outer periphery of the annular ring, wherein when the coupling is in the first position, the protrusion is either within a predetermined distance of the edge of the semi-annular ring or contacts the edge of the semi-annular ring, and wherein when the coupling is in the second position, the protrusion is spaced apart from the edge of the semi-annular ring by a predetermined minimum distance.

10. The release assembly of claim 9, wherein in the pre-installed state, the spring is configured to rotationally bias the coupling about an axis defined by the transverse member towards the first position.

11. The release assembly of claim 10, wherein in the installed state, the spring is configured to bias the distal of the actuation lever towards the lever of the locking device.

12. The release assembly of claim 8, wherein the actuation lever includes a proximal end, a distal end, and a semi-annular ring, wherein the barrel includes an annular ring and a protrusion, wherein the annular ring defines the first receptacle and the protrusion extends from an outer periphery of the annular ring, wherein the semi-annular ring defines the second receptacle and includes a first portion and a second portion, wherein the first portion includes a flange and the second portion extends from the proximal end of the actuation lever and terminates at an edge, wherein as a force is applied to the handle, the transverse member is configured to rotate about a rotational axis to rotate the actuation lever so that the distal end of the actuation lever engages and displaces a locking lever of the locking device to unlock the translatable rail from the fixed rail, and wherein when the coupling is either the first position or the second position, the protrusion of the barrel is disposed between the flange of the semi-annular ring and the proximal portion of the actuation lever, with respect to the rotational axis defined by the transverse member.

13. The release assembly of claim 8, wherein the actuation lever includes a distal end, a proximal end, and a semi-closed cup extending from the proximal end and forming the second receptacle, wherein as a force is applied to the handle, the transverse member is configured to rotate about a rotational axis to rotate the actuation lever so that the distal end of the actuation lever engages and displaces a locking lever of the locking device to unlock the translatable rail from the fixed rail, wherein the semi-closed cup includes a base and a semi-annular ring extending from the base in a direction parallel to the rotational axis, and wherein the base of semi-closed cup defines an aperture that receives an end portion of the transverse member.

14. A method of assembling a release assembly for use in a vehicle seat, the vehicle seat including a pair of rail assemblies, wherein each rail assembly of the pair of rail assemblies includes a fixed rail, a translatable rail, and a locking device, the locking device configured to selectively lock the translatable rail to the fixed rail and unlock the translatable rail from the fixed rail to permit translation of a translatable rail of one rail assembly of the pair of rail assembly along a fixed rail of the one rail assembly in a longitudinal direction, the method comprising: attaching a spring to a coupling, the spring including a first end and a second end, wherein the coupling defines a first receptacle and a channel, wherein the attaching step includes inserting the first end of the spring into the channel defined by the coupling; inserting a portion of the coupling into a second receptacle defined by an actuation lever, wherein the actuation lever is fixed to a transverse member, wherein actuation lever is configured to pivot about an axis defined transverse member to engage a locking lever of the locking device to unlock the translatable rail from the fixed rail; and rotating the coupling in a first rotational direction, relative to the second receptacle, to a first rotational position so that the second end of the spring engages the actuation lever.

15. The method of claim 14, further comprising: rotating the coupling in a second rotational direction to a second rotational position, so that an axial position of coupling is fixed with respect to the axis defined by the transverse member, wherein the second rotational direction is opposite the first rotational direction, and the second rotational position is different than the first rotational position.

16. The method of claim 15, wherein actuation lever includes a proximal end, a distal end, and a semi-annular ring extending from the proximal end and terminating at an edge, wherein the second

receptacle is at least partially defined by the semi-annular ring, wherein the coupling includes an annular ring, a flange and a protrusion, wherein the annular ring forms the first receptacle, wherein the flange extends from the annular ring and the protrusion extends from an outer periphery of the outer ring, and wherein the rotating the coupling in the second rotational direction step includes rotating the coupling so that the protrusion is spaced apart from the edge of the semi-annular ring by a predetermined distance.

17. The method of claim 15, further comprising: fixing the coupling relative to the one translatable rail, wherein the one rail assembly includes a mounting bracket fixed to the translatable rail, and the coupling includes an annular ring and a one or more flanges extending from the annular ring, wherein the annular ring forms the first receptacle, and wherein the fixing step includes attaching the one or more flanges to the mounting bracket.

18. The method of claim 17, further comprising: rotating the coupling in the first rotational direction relative to the second receptacle to a third rotational position, wherein the fixing step and the rotating the coupling to the third rotational position step are accomplished substantially simultaneously.

19. The method of claim 17, wherein the mounting bracket includes one or more locking protrusions, wherein the one or more flanges define one or more apertures, and wherein the fixing step includes placing the flange of the coupling onto the mounting bracket and the one or more apertures receiving the one or more locking protrusions.

20. The method of claim 14, further comprising: inserting an end region of the transverse member into a third receptacle, wherein the coupling includes a barrel and an annular ring extending from the barrel, wherein the barrel forms a third receptacle.
